From the $C_1 \otimes B_2$ row to §4.2 of Lublinsky–Mulian

The p^+ integration done term by term — UV, finite and rapidity terms, all of them

arXiv:1610.03453v2, §4.2 and Appendices C, H.2 — one continuous derivation

0. The short version

Your row *is* the vertex of Eq. (4.16) of the paper. Under the identification

$$\boxed{z_{\text{paper}} \leftrightarrow x, \qquad z'_{\text{paper}} \leftrightarrow z, \qquad y \leftrightarrow y}$$

one has $Y = y - z_{\text{paper}} = y - x = s$, $Y' = y - z'_{\text{paper}} = y - z = s - r$, $Z = z_{\text{paper}} - z'_{\text{paper}} = x - z = -r$, and therefore

$$\bar{\xi}\, D \;=\; \bar{\xi}\,(s-r)^2 + \xi\, s^2 \;=\; (1-\xi)(Y')^2 + \xi\, Y^2,$$

which is *exactly* the second energy denominator of (4.16). Better still, the six ξ–weights of your bracket are exactly the six products that the paper's vertex Λ, Θ decomposition produces. So the whole of §4.2 / Appendix H.2 applies, and the p^+ integral is elementary: it needs only the paper's (C.2), (C.5), (C.6). Doing it gives

$$\int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi\, \frac{\text{bracket}}{\bar{\xi} D} \;=\; \underbrace{\ell_k \left[\frac{V_2}{Y^2} + \frac{V_4}{(Y')^2}\right]}_{\text{(a) rapidity} \to (\delta Y)^2\text{, Eq. (H.7)}} \;+\; \underbrace{\ln\frac{Y^2}{(Y')^2}\, \mathcal{G}_{\text{log}}}_{\text{(b) the NLO logs, Eq. (H.6)}} \;+\; \underbrace{\mathcal{G}_{\text{rat}}}_{\text{(c) rational, Eq. (H.6)}},$$

with $\ell_k \equiv \ln(k^+/\Lambda)$, and the ultraviolet ($z \to x$) content of the whole thing sitting in one place:

$$\left[\int d\xi\, \frac{\text{bracket}}{\bar{\xi} D}\right]_{r\to 0} = \left(2\ell_k - \frac{11}{6}\right) \frac{(s\cdot P)}{P^2 s^2}\, \frac{1}{r^2} + (\text{integrable}), \qquad 2\ell_k - \frac{11}{6} = \int d\xi\, \frac{P_{gg}(\xi)}{2N_c}.$$

The $2\ell_k$ comes entirely from block (a) and the $-\frac{11}{6}$ entirely from blocks (b)+(c) — verified numerically in §10E. And $-\frac{11}{6} \times 2N_c = -\frac{11N_c}{3} = -b$ at $N_f = 0$: this row is a running–coupling contribution. §9 shows that its ξ integral is *literally* the paper's (C.10), which is where $\frac{67}{18} - \frac{\pi^2}{3}$ comes from.

1. Three things in the expression as typed

These have to be settled before any integral, because two of them change the answer.

(i) Θ_6 carries a minus sign twice. In (1.1) the last structure already appears with an overall minus, and your (1.10) gives the *result* of contracting it:

$$-\Big[\tfrac{(y-x)^m(x-z)^i}{(x-z)^2}-\tfrac{(y-x)^m(y-z)^i}{2(y-z)^2}+\tfrac{(y-z)^m(x-z)^i}{(x-z)^2}+\tfrac{(y-x)^i(y-z)^m}{2(y-x)^2}\Big]\frac{P^i r^m}{P^2r^2}=+\frac{2(P\cdot r)(r\cdot s)}{P^2r^4}-\frac{P\cdot r}{P^2r^2}+\dots$$

In the expression you typed this same right–hand side appears again behind a minus sign, $-\big[\frac{2(P\cdot r)(r\cdot s)}{P^2r^4}-\dots\big]$. That flips Θ_6. It matters: with the sign of (1.10) the coefficient of $\frac{(P\cdot r)(s\cdot r)}{r^4}$ is $d_\perp\xi\bar\xi-2\bar\xi-2\xi+2=d_\perp\xi\bar\xi$ and the UV coefficient comes out as $P_{gg}/2N_c$; with the extra minus it is $d_\perp\xi\bar\xi-4$ and P_{gg} never appears. The sign of (1.10) is used below.

(ii) Θ_4: the factor 2, again. Contracting the Θ_4 structure of (1.1), $-\frac{k^+-p^+}{p^+}\big[\frac{(y-z)^i(x-z)^m}{(x-z)^2}+\frac{(y-x)^m(y-z)^i}{2(y-x)^2}\big]$, with $\frac{P^ir^m}{P^2r^2}$ gives $\frac{\bar\xi}{\xi}\frac{(s\cdot P)-(P\cdot r)}{P^2r^2}\big[1-\frac{r\cdot s}{2s^2}\big]$ — the $2s^2$ that is missing in (1.8) and in the expression you typed. Since you have now written it twice without the 2, everything below is carried with a parameter

$$V_4=\frac{(s\cdot P)-(P\cdot r)}{P^2r^2}\Big[1-\kappa\,\frac{r\cdot s}{s^2}\Big],\qquad \kappa=\tfrac12\ (\text{from (1.1)}),\qquad \kappa=1\ (\text{as printed}).$$

Every result is given for both. Nothing that matters here depends on it: κ does not enter the UV coefficient, nor the rapidity block, nor the ξ integrals — only the finite remainder (§10D shows the UV coefficient is κ–independent to 7 digits).

(iii) The phase. Written out, $e^{-ik\cdot(w'-z)}e^{-ik\cdot(z-x)}=e^{-ik\cdot w'}e^{ik\cdot x}$ has no r–dependence at all, so (1.3) is consistent only if q is treated as ξ–independent, which is how (1.11) is written ($\int d\xi$ outside, $e^{iq\cdot r}$ inside). That is what is done below, and it is also what the paper needs: §4.2 has *one* power of $\ln(\Lambda/k^+)$ and hence $(\delta Y)^2$. If instead $q=\bar\xi k$ (the second exponential being $e^{-i\xi k\cdot(z-x)}$), the ξ integrals below acquire a factor $e^{i\bar\xi k\cdot r}$: the ℓ_k terms are unchanged (they come from the endpoints, where the phase is $e^{ik\cdot r}$ and 1), but the finite ones stop being elementary and an extra $\ln(1/\bar\xi)$ appears, generating ℓ_k^2. See §9.

2. Variables, and the map to the paper

Keep your notation, $r=z-x$, $s=y-x$, $P=x'-w'$, $\xi=p^+/k^+$, $\bar\xi=1-\xi$. Then, as in (1.2),

$$p^+(y-x)^2+(k^+-p^+)(y-z)^2=k^+\big[\xi s^2+\bar\xi(s-r)^2\big]=k^+\bar\xi D,\qquad D=(r-s)^2+M^2,\quad M^2=\frac{\xi}{\bar\xi}s^2.$$

Now put the paper's variables beside them, using $z_{\text{paper}}=x$, $z'_{\text{paper}}=z$:

paper (§2.8, §4.2)	here	paper	here
$Y=y-z$	$y-x=s$	Y^2	$s^2\equiv B$
$Y'=y-z'$	$y-z=s-r$	$(Y')^2$	$(s-r)^2\equiv A$
$Z=z-z'$	$x-z=-r$	Z^2	r^2
$Y^2-(Y')^2$	$2s\cdot r-r^2\equiv C$	$\ln\frac{Y^2}{(Y')^2}$	$L\equiv\ln\frac{s^2}{(s-r)^2}$

$$\bar\xi\,D\;=\;\bar\xi\,(Y')^2+\xi\,Y^2\;=\;\Delta(\xi)\;=\;A+\xi\,C,\qquad C=B-A.$$

This is the denominator $\big((1-\xi)(Y')^2+\xi Y^2\big)$ of Eq. (4.16). Your $1/(\bar\xi D)$ and the paper's second energy denominator are the same object.

3. The bracket is the paper's vertex, contracted with one LO kernel

Strip the ξ–dependence off your six terms. Writing the bracket of your expression as

$$\text{bracket}\;=\;\xi\bar\xi\,V_1+\frac{\xi}{\bar\xi}V_2+\bar\xi\,V_3+\frac{\bar\xi}{\xi}V_4+\xi\,V_5+V_6,$$

the six V_i are exactly your (1.4)–(1.10) with the prefactor removed:

$$V_1 = \frac{d_\perp}{2}\left[\frac{2(P\cdot r)(s\cdot r)}{P^2r^4} - \frac{P\cdot r}{P^2r^2}\right], \qquad V_2 = \frac{P\cdot s}{P^2r^2} + \frac{(P\cdot s)(r\cdot s - r^2)}{2P^2r^2(s-r)^2},$$
$$V_3 = \frac{P\cdot r}{P^2r^2}\left[2 - \frac{2(s\cdot r)}{r^2} - \frac{s\cdot r}{2s^2}\right], \qquad V_4 = \frac{(s\cdot P)-(P\cdot r)}{P^2r^2}\left[1 - \kappa\frac{r\cdot s}{s^2}\right],$$
$$V_5 = -\frac{P\cdot r}{P^2}\left[\frac{2(s\cdot r)}{r^4} + \frac{s^2 - s\cdot r}{2r^2(s-r)^2} - \frac{1}{2r^2}\right], \quad V_6 = \frac{2(P\cdot r)(r\cdot s)}{P^2r^4} - \frac{P\cdot r}{P^2r^2} + \frac{P\cdot s}{2P^2s^2}$$
$$\qquad - \frac{(P\cdot s)(r\cdot s)}{2P^2r^2s^2} + \frac{(r\cdot s)\big[(P\cdot s)-(P\cdot r)\big]}{2P^2r^2(s-r)^2}.$$

(That this reproduces the contraction of (1.1) exactly, with $\kappa = \frac{1}{2}$ and with the V_6 sign of (1.10), is checked symbolically in §10.)

Now compare with the paper. The vertex of (4.16), second factor, is

$$\underbrace{\frac{Y^2-(Y')^2}{2Z^2}\,\delta^{jl}}_{\Theta^{jl}(y,z,z')} + \frac{1}{\xi}\underbrace{\left[\frac{(Y')^jZ^l}{Z^2} + \frac{Y^l(Y')^j}{2Y^2}\right]}_{\Lambda^{jl}(y,z,z')} + \frac{1}{1-\xi}\underbrace{\left[\frac{Y^lZ^j}{Z^2} - \frac{Y^l(Y')^j}{2(Y')^2}\right]}_{\Lambda^{lj}(y,z',z)\text{ with } z\leftrightarrow z'},$$

in the notation (H.8). With $Y = s$, $Y' = s - r$, $Z = -r$ these are, structure for structure, your Θ_1, your Θ_4 and your Θ_2:

$$\Lambda^{jl}(y,z,z') \;\longleftrightarrow\; \frac{(s-r)^i(-r)^m}{r^2} + \frac{s^m(s-r)^i}{2s^2} \qquad (\text{the bracket of your } \Theta_4),$$
$$\Lambda^{lj}(y,z',z) \;\longleftrightarrow\; \frac{s^i(-r)^m}{r^2} - \frac{s^i(s-r)^m}{2(s-r)^2} \qquad (\text{the bracket of your } \Theta_2).$$

The difference between your row and (4.16) is only *what the vertex is contracted with*. The paper contracts vertex × vertex, so the six products of $\{\Theta, \frac{1}{\xi}\Lambda, \frac{1}{1-\xi}\Lambda\}$ with themselves carry the weights $\xi\bar{\xi}\cdot\{1,\frac{1}{\xi},\frac{1}{\bar{\xi}}\}\otimes\{1,\frac{1}{\xi},\frac{1}{\bar{\xi}}\}$; your row contracts the vertex with the LO Weizsäcker–Williams kernel $\frac{(x'-w')^i(z-x)^m}{(x'-w')^2(z-x)^2} = \frac{P^ir^m}{P^2r^2}$, and the six weights collapse to

$$\Big\{\ \xi\bar{\xi},\ \ \frac{\xi}{\bar{\xi}},\ \ \bar{\xi},\ \ \frac{\bar{\xi}}{\xi},\ \ \xi,\ \ 1\ \Big\} \;=\; \xi\bar{\xi}\cdot\Big\{1,\tfrac{1}{\bar{\xi}^2},\tfrac{1}{\xi},\tfrac{1}{\xi^2},\tfrac{1}{\bar{\xi}},\tfrac{1}{\xi\bar{\xi}}\Big\},$$

which is precisely the set that appears in (4.16) — this is the check that your row and the paper's are built from the same vertex. It is also why the p^+ integral needs nothing beyond (C.2), (C.5) and (C.6).

4. The p^+ integration: six elementary integrals

Everything ξ–dependent is now in the weights and in $\Delta(\xi) = A + \xi C$. Define, with $\lambda = \Lambda/k^+$, $\ell_k = \ln(1/\lambda) = \ln(k^+/\Lambda)$,

$$J_i \;\equiv\; \int_\lambda^{1-\lambda} d\xi\,\frac{w_i(\xi)}{\Delta(\xi)}, \qquad w = \Big(\xi\bar{\xi},\ \frac{\xi}{\bar{\xi}},\ \bar{\xi},\ \frac{\bar{\xi}}{\xi},\ \xi,\ 1\Big),$$

so that $\displaystyle\int d\xi\,\frac{\text{bracket}}{\bar{\xi}D} = \sum_{i=1}^{6} J_i\,V_i$. Derive them in the order in which they build on one another. Everything follows from

$$\frac{1}{\Delta} = \frac{1}{A+\xi C}, \qquad \frac{1}{\xi\Delta} = \frac{1}{A}\Big[\frac{1}{\xi} - \frac{C}{\Delta}\Big], \qquad \frac{1}{\bar{\xi}\Delta} = \frac{1}{B}\Big[\frac{1}{\bar{\xi}} + \frac{C}{\Delta}\Big], \qquad \xi = \frac{\Delta - A}{C}.$$

J_6 (no weight). $\int_0^1 d\xi/\Delta = \frac{1}{C}\ln\frac{B}{A}$ — this is the paper's (C.2). Write $L \equiv \ln(B/A) = \ln\big[Y^2/(Y')^2\big]$:

$$J_6 = \frac{L}{C}.$$

J_5 ($w = \xi$). Using $\xi = (\Delta - A)/C$: $\int\frac{\xi}{\Delta} = \frac{1}{C}\int\big(1 - \frac{A}{\Delta}\big) = \frac{1}{C} - \frac{A}{C}J_6$, i.e.

$$J_5 = \frac{1}{C} - \frac{A}{C^2}L.$$

J_3 ($w = \bar{\xi}$). Simply $J_3 = J_6 - J_5$, and $1 + \frac{A}{C} = \frac{B}{C}$:

$$J_3 = \frac{B}{C^2}L - \frac{1}{C}.$$

J_1 ($w = \xi\bar\xi$). $J_1 = J_5 - \int \frac{\xi^2}{\Delta}$, and with $\xi^2 = \frac{(\Delta-A)^2}{C^2}$, $\int_0^1 \Delta\, d\xi = A + \frac{C}{2}$: $\int \frac{\xi^2}{\Delta} = \frac{1}{C^2}\left[\frac{C}{2} - A + \frac{A^2}{C} L\right]$, so

$$J_1 = \frac{1}{2C} + \frac{A}{C^2} - \frac{AB}{C^3}\, L.$$

J_4 ($w = \bar\xi/\xi$) — the first place a rapidity logarithm appears. From $\frac{\bar\xi}{\xi} = \frac{1}{\xi} - 1$ and the partial fraction above, $\int_\lambda^{1-\lambda} \frac{d\xi}{\xi\Delta} = \frac{1}{A}\left[\ell_k - L\right]$ — this is the paper's (C.5). Subtracting J_6 and using $\frac{1}{A} + \frac{1}{C} = \frac{B}{AC}$:

$$J_4 = \frac{\ell_k}{A} - \frac{B}{AC}\, L.$$

J_2 ($w = \xi/\bar\xi$) — the second rapidity logarithm. With $t = \bar\xi$, $\Delta = B - tC$, $\frac{1}{t(B-tC)} = \frac{1}{B}\left[\frac{1}{t} + \frac{C}{B-tC}\right]$, giving $\int_\lambda^{1-\lambda} \frac{d\xi}{\bar\xi\Delta} = \frac{1}{B}\left[\ell_k + L\right]$. Subtracting J_6 and using $\frac{1}{B} - \frac{1}{C} = -\frac{A}{BC}$:

$$J_2 = \frac{\ell_k}{B} - \frac{A}{BC}\, L.$$

> **A sign in the paper.** $\int \frac{d\xi}{\bar\xi\Delta} = \frac{1}{B}(\ell_k + L)$ is the paper's (C.6), which is printed as $-\frac{1}{W^2}\left(\ln\frac{\Lambda}{k^+} + \ln\frac{W^2}{(W')^2}\right) = \frac{1}{B}(\ell_k - L)$. The L should come with the opposite sign: (C.6) is (C.5) with $\xi \to 1-\xi$, which swaps $A \leftrightarrow B$ and hence $L \to -L$, turning $\frac{1}{A}(\ell_k - L)$ into $\frac{1}{B}(\ell_k + L)$. Direct quadrature confirms it (§10B): for $A = 1, B = 4$ the integral is 5.52739, not 4.83424. (C.5) itself is right.

5. The result of the p^+ integration

$$\mathcal{G}(r) \;\equiv\; \int_{\Lambda/k^+}^{1-\Lambda/k^+} d\xi\, \frac{\text{bracket}}{\bar\xi D} \;=\; \sum_{i=1}^{6} J_i\, V_i,$$

$$J_1 = \frac{1}{2C} + \frac{A}{C^2} - \frac{AB}{C^3}L, \quad J_2 = \frac{\ell_k}{B} - \frac{A}{BC}L, \quad J_3 = \frac{B}{C^2}L - \frac{1}{C},$$
$$J_4 = \frac{\ell_k}{A} - \frac{B}{AC}L, \quad J_5 = \frac{1}{C} - \frac{A}{C^2}L, \quad J_6 = \frac{L}{C},$$

$$A = (s-r)^2 = (Y')^2, \qquad B = s^2 = Y^2, \qquad C = B - A = 2s\cdot r - r^2, \qquad L = \ln\frac{B}{A}, \qquad \ell_k = \ln\frac{k^+}{\Lambda}.$$

Regrouped by the object that multiplies each piece — which is how the paper presents it —

$$\mathcal{G} = \ell_k\, \mathcal{G}_{\rm rap} + L\, \mathcal{G}_{\rm log} + \mathcal{G}_{\rm rat},$$
$$\mathcal{G}_{\rm rap} = \frac{V_2}{B} + \frac{V_4}{A} = \frac{V_2}{Y^2} + \frac{V_4}{(Y')^2},$$
$$\mathcal{G}_{\rm log} = -\frac{AB}{C^3}V_1 - \frac{A}{BC}V_2 + \frac{B}{C^2}V_3 - \frac{B}{AC}V_4 - \frac{A}{C^2}V_5 + \frac{1}{C}V_6,$$
$$\mathcal{G}_{\rm rat} = \left(\frac{1}{2C} + \frac{A}{C^2}\right)V_1 - \frac{1}{C}V_3 + \frac{1}{C}V_5.$$

5.1 What each block is, in the paper's language

$\ell_k\mathcal{G}_{\rm rap}$ **is Eq. (H.7).** The paper's $(\delta Y)^2$ piece is

$$\Sigma^{(\delta Y)^2}_{JSSJ} \;\propto\; \left(\frac{\operatorname{tr}\left(\Lambda(x,z,z')\Lambda^T(y,z,z')\right)}{(X')^2(Y')^2} + \frac{\operatorname{tr}\left(\Lambda(x,z',z)\Lambda^T(y,z',z)\right)}{X^2Y^2}\right)\ln\frac{\Lambda}{k^+},$$

i.e. the $1/\xi$ vertex divided by $(Y')^2$ plus the $1/(1-\xi)$ vertex divided by Y^2. Ours is the same thing with one vertex instead of two:

$$\mathcal{G}_{\rm rap} = \underbrace{\frac{V_4}{(Y')^2}}_{1/\xi\text{ vertex}} + \underbrace{\frac{V_2}{Y^2}}_{1/(1-\xi)\text{ vertex}} = \frac{(s\cdot P) - (P\cdot r)}{P^2r^2(s-r)^2}\left[1 - \kappa\frac{r\cdot s}{s^2}\right] + \frac{P\cdot s}{P^2r^2s^2} + \frac{(P\cdot s)(r\cdot s - r^2)}{2P^2r^2(s-r)^2s^2}.$$

The correspondence is exact, term for term: it is the $1/\xi$ and $1/(1-\xi)$ poles of the vertex that produce ℓ_k, and each is divided by the value of $\Delta(\xi)$ at the endpoint where its pole sits — $\Delta(0) = A = (Y')^2$ for the $1/\xi$ pole, $\Delta(1) = B = Y^2$ for the $1/(1-\xi)$ pole.

$L\,\mathcal{G}_{\rm log} + \mathcal{G}_{\rm rat}$ **is Eq. (H.6).** (H.6) has the same shape: a rational term $(-\frac{1}{2Z^4})$ plus terms carrying $\ln\frac{X^2}{(X')^2}$ and $\ln\frac{Y^2}{(Y')^2}$. Ours has one such logarithm, $L = \ln\frac{Y^2}{(Y')^2}$, because there is one energy denominator.

> **Use $\sum_i J_i V_i$, not the regrouping, for numbers.** $\mathcal{G}_{\rm log}$ and $\mathcal{G}_{\rm rat}$ have spurious $1/C^3$ poles that cancel between them ($C \to 0$ is the circle $|r-s| = |s|$, where nothing happens physically). Each J_i separately is regular there, with
>
> $$J_i = \frac{1}{B}\sum_{n\geq 0}\left(\frac{C}{B}\right)^n \int_\lambda^{1-\lambda} d\xi\, w_i(\xi)\bar{\xi}^n \implies \begin{array}{lll} J_1 B = \frac{1}{6} + \frac{\epsilon}{12} + \frac{\epsilon^2}{20} + \dots & J_2 B = (\ell_k - 1) + \frac{\epsilon}{2} + \dots & J_3 B = \frac{1}{2} + \frac{\epsilon}{3} + \dots \\ J_4 B = (\ell_k - 1) + \epsilon(\ell_k - \frac{3}{2}) + \dots & J_5 B = \frac{1}{2} + \frac{\epsilon}{6} + \dots & J_6 B = 1 + \frac{\epsilon}{2} + \dots \end{array}$$
>
> with $\epsilon = C/B$. Evaluating J_1 from the closed form at $\epsilon \lesssim 10^{-5}$ loses all significance (§10G); use the series there.

6. The k^+ integration — δY and $(\delta Y)^2$

Your expression is differential in k^+. To land on the form of §4.2, integrate over the rapidity slice as the paper does, $\int_\Lambda^{e^{\delta Y}\Lambda}\frac{dk^+}{k^+}$:

$$\int_\Lambda^{e^{\delta Y}\Lambda}\frac{dk^+}{k^+} = \delta Y, \qquad \int_\Lambda^{e^{\delta Y}\Lambda}\frac{dk^+}{k^+}\,\ell_k = \int_\Lambda^{e^{\delta Y}\Lambda}\frac{dk^+}{k^+}\ln\frac{k^+}{\Lambda} = \frac{(\delta Y)^2}{2},$$

so that, exactly as in (4.18),

$$\int\frac{dk^+}{k^+}\,\mathcal{G} \;=\; \underbrace{\delta Y\Big[L\,\mathcal{G}_{\rm log} + \mathcal{G}_{\rm rat}\Big]}_{\text{the NLO kernel, cf. (4.19), (H.6)}} \;+\; \underbrace{\frac{(\delta Y)^2}{2}\,\mathcal{G}_{\rm rap}}_{\text{cf. (4.20), (H.7)}}\,.$$

The $(\delta Y)^2$ piece is the one that must reassemble into $\frac{1}{2}(\delta Y\, H^{\rm LO}_{\rm JIMWLK})^2$, Eq. (4.11) — and indeed $\mathcal{G}_{\rm rap}$ is built only out of products of LO Weizsäcker–Williams kernels $\frac{r^i}{r^2}$, $\frac{s^i}{s^2}$, $\frac{(s-r)^i}{(s-r)^2}$, $\frac{P^i}{P^2}$, which is what (4.20) and (4.38) are made of.

7. UV, finite, and rapidity terms

Now separate. The only divergence left in $\mathcal{G}$ is at $r \to 0$, i.e. $z \to x$, where $\int d^2z$ hits $1/r^2$. Every J_i is regular there ($C \to 0$), so the UV coefficient is $J_i(r=0)$ times the $r \to 0$ behaviour of V_i. In two dimensions $\langle \hat{r}^i\rangle = 0$, $\langle \hat{r}^i\hat{r}^j\rangle = \frac{1}{2}\delta^{ij}$, hence $\left\langle\frac{(P\cdot r)(s\cdot r)}{r^4}\right\rangle = \frac{s\cdot P}{2r^2}$ and $\left\langle\frac{P\cdot r}{r^2}\right\rangle = 0$:

	weight w_i	$V_i\big\vert_{r\to 0}$ **in units of** $\frac{s\cdot P}{P^2 r^2}$	$B\,J_i(r{=}0) = \int d\xi\, w_i$	**product**
V_1 (Θ_1, instantaneous)	$\xi\bar{\xi}$	$+\frac{d_\perp}{2} = +1$	$\frac{1}{6}$	$+\frac{1}{6}$
V_2 (Θ_2)	$\xi/\bar{\xi}$	$+1$	$\ell_k - 1$	$\ell_k - 1$
V_3 (Θ_3)	$\bar{\xi}$	-1	$\frac{1}{2}$	$-\frac{1}{2}$
V_4 (Θ_4)	$\bar{\xi}/\xi$	$+1$	$\ell_k - 1$	$\ell_k - 1$
V_5 (Θ_5)	ξ	-1	$\frac{1}{2}$	$-\frac{1}{2}$
V_6 (Θ_6)	1	$+1$	1	$+1$
total		$C_{\rm UV}(\xi) = \xi\bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi}$		$2\ell_k - \frac{11}{6}$

Two things to read off the table. Before the ξ integral, the $r \to 0$ coefficient is $\xi\bar{\xi}\cdot 1 + \frac{\xi}{\bar{\xi}}\cdot 1 + \bar{\xi}(-1) + \frac{\bar{\xi}}{\xi}\cdot 1 + \xi(-1) + 1 = \xi\bar{\xi} + \frac{\xi}{\bar{\xi}} + \frac{\bar{\xi}}{\xi} = \frac{P_{gg}(\xi)}{2N_c}$, because $-\bar{\xi} - \xi + 1 = 0$: $\Theta_3, \Theta_5, \Theta_6$ cancel among themselves in the UV, and $\xi(1-\xi)$ comes from Θ_1, $\frac{\xi}{1-\xi}$ from Θ_2, $\frac{1-\xi}{\xi}$ from Θ_4. After the ξ integral the same cancellation leaves $2\ell_k - \frac{11}{6}$. κ never enters (§10D).

The separation. Exactly, for all r:

$$\mathcal{G}(r) \;=\; \underbrace{\Big(2\ell_k-\frac{11}{6}\Big)\frac{s\cdot P}{P^2s^2}\,\frac{1}{r^2}}_{\textbf{UV}\ (z\to x)} \;+\; \underbrace{\ell_k\Big[\mathcal{G}_{\rm rap}-\frac{2}{s^2}\,\frac{s\cdot P}{P^2r^2}\Big]}_{\textbf{rapidity, finite at } z\to x} \;+\; \underbrace{\Big[L\,\mathcal{G}_{\log}+\mathcal{G}_{\rm rat}+\frac{11}{6}\,\frac{s\cdot P}{P^2s^2r^2}\Big]}_{\textbf{finite}},$$

the last two blocks being integrable at $r=0$. The $2\ell_k$ of the UV coefficient comes entirely from the rapidity block and the $-\frac{11}{6}$ entirely from the other two; both are confirmed to six digits in §10E.

Only the first block needs a regulator. Doing its z integral, $\int d^2r\, e^{iq\cdot r}/(r^2+m^2)=2\pi K_0(m|q|)=\pi\log\frac{4e^{-2\gamma_E}}{m^2q^2}$, gives

$$\Big(2\ell_k-\frac{11}{6}\Big)\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\,\pi\log\frac{4e^{-2\gamma_E}}{m^2q^2}, \qquad -\frac{11}{6}\times 2N_c=-\frac{11N_c}{3}=-b\big|_{N_f=0},$$

a product of two Weizsäcker–Williams kernels times $b\log$ — the running–coupling structure of K_{JSJ}, Eq. (2.64)/(2.66).

8. The other order, and where $\frac{67}{18}-\frac{\pi^2}{3}$ comes from

If instead the z integral is done first (the companion note), the UV logarithm comes out carrying a $\bar\xi$ inside its argument, because $D_0\equiv D|_{r=0}=s^2/\bar\xi$:

$$\frac{1}{\bar\xi}\,\frac{\mathcal{W}_{\rm UV}}{P^2}=\pi\,\frac{P_{gg}(\xi)}{2N_c}\,\frac{(x'-w')\cdot(y-x)}{(x'-w')^2(y-x)^2}\,\log\frac{(y-x)^2}{\bar\xi\,m^2},$$

and the ξ integral then needs the logarithmic moments of P_{gg}. These are exactly the paper's (C.10). Writing $C_{\rm UV}=\xi\bar\xi+\frac{\xi}{\bar\xi}+\frac{\bar\xi}{\xi}=\xi(1-\xi)-2+\frac{1}{\xi(1-\xi)}$ — which is *literally* the weight of (C.10) —

$$\int_\lambda^{1-\lambda} d\xi\, C_{\rm UV}=2\ell_k-\frac{11}{6}, \qquad \int_\lambda^{1-\lambda} d\xi\, C_{\rm UV}\,\ln\big[\xi(1-\xi)\big]=-\ell_k^2+\frac{67}{18}-\frac{\pi^2}{3},$$

the second being (C.10) with the $\ln(k^2/\mu^2)$ piece removed. Since $C_{\rm UV}$ is symmetric under $\xi\leftrightarrow\bar\xi$, $\int C_{\rm UV}\ln(\xi\bar\xi)=2\int C_{\rm UV}\ln\bar\xi$, so the moment this row actually needs is half of it:

$$\int_\lambda^{1-\lambda} d\xi\; C_{\rm UV}(\xi)\,\log\frac{1}{1-\xi}=\frac{\ell_k^2}{2}+\frac{\pi^2}{6}-\frac{67}{36}=-\frac{1}{2}\left(-\ell_k^2+\frac{67}{18}-\frac{\pi^2}{3}\right),$$

giving, for the whole UV block of this row,

$$\int\frac{dp^+}{2\pi}\,\frac{P_{gg}(\xi)}{2N_c}\,\log\frac{k^+(y-x)^2}{(k^+-p^+)m^2}=\frac{k^+}{2\pi}\left[\Big(2\ell_k-\frac{11}{6}\Big)\log\frac{(y-x)^2}{m^2}+\frac{\ell_k^2}{2}+\frac{\pi^2}{6}-\frac{67}{36}\right].$$

So: $-\frac{11}{6}\times 2N_c=-b$ at $N_f=0$, and the constant is exactly the $\frac{67}{18}-\frac{\pi^2}{3}$ of (C.10) — the same combination that ends up as the $\big(\frac{67}{9}-\frac{\pi^2}{3}\big)N_c$ of K_{JSJ}, Eqs. (2.64), (2.66), (4.36). (The remaining factor–of–two bookkeeping between $\frac{67}{18}$ and $\frac{67}{9}$ is the $x\leftrightarrow y$ symmetrisation of (4.31); the π^2 term additionally receives the self–energy/normalisation pieces of (4.31), which are not in this row, so this is an identification of the source of the constant, not a complete assembly of K_{JSJ}.)

> **Why the two orders look different.** Doing ξ first (§5) the UV logarithm is $\log\frac{4e^{-2\gamma_E}}{m^2q^2}$ with coefficient $2\ell_k-\frac{11}{6}$ and there is no ℓ_k^2; doing z first the logarithm is $\log\frac{(y-x)^2}{\xi m^2}$ and the extra $\log\frac{1}{\xi}$ generates $\frac{1}{2}\ell_k^2$. The totals agree; the $\bar\xi$ simply sits in a different block. The coefficient of $\log(1/m^2)$ — the only regulator–dependent number — is $2\ell_k-\frac{11}{6}$ in both.

9. What this reproduces, and what it cannot

Reproduced. The vertex and its six ξ–weights; the energy denominator $(1-\xi)(Y')^2+\xi Y^2$; the p^+ integration using nothing but (C.2), (C.5), (C.6); the resulting three–block structure (rapidity $\times\ln\frac{\Lambda}{k^+}$, a $\ln\frac{Y^2}{(Y')^2}$ block, a rational block) which is the shape of (H.6)–(H.7); the $\ell_k\to\frac{1}{2}(\delta Y)^2$ mechanism of (4.18)–(4.20); the fact that the $(\delta Y)^2$ block is built only from products of LO WW kernels, as (4.20) and (4.38) are; b and $\frac{67}{18}-\frac{\pi^2}{3}$.

Not reproducible from this row, and worth knowing before you spend more time on it. Σ_{JSSJ}, Eq. (4.16), has *two* energy denominators, because both sides of the overlap are the two–gluon wave function $|\psi_{gg\rho}\rangle$. That is why H.2 needs

(C.3) and (C.4) as well, why both $\ln \frac{X^2}{(X')^2}$ and $\ln \frac{Y^2}{(Y')^2}$ appear, and where the $1/Z^4$ of K_{JSSJ} comes from. Your row has *one* energy denominator, one LO Weizsäcker–Williams kernel $\frac{P^i r^m}{P^2 r^2}$ in place of the second vertex, a Fourier phase, and a fifth point w' — it is a production object, d^3N/d^3k, not an S–matrix overlap. It therefore cannot give K_{JSSJ} directly.

The row it actually resembles is **§4.3, Σ_{JJSSJ}, Eq. (4.21)**: one energy denominator, the same vertex, contracted with products of LO kernels. Its ξ integration (4.22) is done with exactly (C.2), (C.5), (C.6) and yields exactly the structure obtained in §5 — a $\ln \frac{W^2}{(W')^2}$ block plus a $\ln \frac{\Lambda}{k^+}$ block whose coefficient is the $1/\xi$ vertex piece divided by $(W')^2$ plus the $1/(1-\xi)$ piece divided by W^2, which is our $\mathcal{G}_{\rm rap} = V_4/(Y')^2 + V_2/Y^2$. If the target is a Σ_{JSSJ}–type kernel, the missing ingredient is the second energy denominator, i.e. the second $|\psi_{gg\rho}\rangle$ — not anything in the ξ or z integration.

10. Checks

All algebra symbolic (sympy) against the contraction of Eq. (1.1) itself; all integrals against quadrature.

```
kinematics: s = [ 0.7 -0.4]  P = [1.3 0.5]   s^2 = 0.6500  s.P = 0.7100  l_k = ln(k+/Lam) = 20.723266

A. the six xi-integrals J_i  (analytic vs quadrature, lambda = 1e-9)
   A=1.0 B=4.0
      J1  analytic +0.072400835390   quad +0.072400835390   rel 7.7e-16
      J2  analytic +5.065291929143   quad +5.065291949611   rel 4.0e-09
      J3  analytic +0.282797493831   quad +0.282797492831   rel 3.5e-09
      J4  analytic +18.874873355453   quad +18.874873359453   rel 2.1e-10
      J5  analytic +0.179300626542   quad +0.179300626292   rel 1.4e-09
      J6  analytic +0.462098120373   quad +0.462098119123   rel 2.7e-09
   A=2.3 B=0.7
      J1  analytic +0.118351965902   quad +0.118351965902   rel 5.9e-16
      J2  analytic +27.161769629736   quad +27.161769751034   rel 4.5e-09
      J3  analytic +0.299723106714   quad +0.299723106279   rel 1.5e-09
      J4  analytic +8.783836003343   quad +8.783836003475   rel 1.5e-11
      J5  analytic +0.443766935082   quad +0.443766933653   rel 3.2e-09
      J6  analytic +0.743490041796   quad +0.743490039933   rel 2.5e-09
   A=0.9 B=1.1
      J1  analytic +0.167001436559   quad +0.167001436559   rel 9.0e-15
      J2  analytic +18.018407006697   quad +18.018407081643   rel 4.2e-09
      J3  analytic +0.518444125209   quad +0.518444124098   rel 2.1e-09
      J4  analytic +21.799530013227   quad +21.799530014585   rel 6.2e-11
      J5  analytic +0.484909352102   quad +0.484909351193   rel 1.9e-09
      J6  analytic +1.003353477311   quad +1.003353475291   rel 2.0e-09

B. the paper's (C.5) and (C.6)      [A=(W')^2=1, B=W^2=4]
   (C.5) quadrature            19.3369714786
         paper  (l_k - L)/A  = 19.3369714758     agrees
   (C.6) quadrature            5.5273900687
         paper  (l_k - L)/B  = 4.8342428690     DISAGREES  (sign of L)
         correct (l_k + L)/B = 5.5273900495     agrees

C. the kernel G(r) = sum_i J_i V_i   vs direct xi-quadrature of the whole bracket
   r=[0.31 0.22] kappa=0.5: assembled +99.813352876  quad +99.813353193  rel 3.2e-09
   r=[0.31 0.22] kappa=1.0: assembled +97.262612502  quad +97.262612819  rel 3.3e-09
   r=[-0.5  0.8] kappa=0.5: assembled +14.920327198  quad +14.920327236  rel 2.6e-09
   r=[-0.5  0.8] kappa=1.0: assembled +16.939098704  quad +16.939098743  rel 2.3e-09
   r=[ 0.12 -0.05] kappa=0.5: assembled +1431.676306092  quad +1431.676309142  rel 2.1e-09
   r=[ 0.12 -0.05] kappa=1.0: assembled +1371.521965492  quad +1371.521968538  rel 2.2e-09

D. UV coefficient of each vertex V_i  (units of (s.P)/(P^2 r^2), angular average at r->0)
   kappa=0.5 : [ 1.  1. -1.  1. -1.  1.]
   kappa=1.0 : [ 1.  1. -1.  1. -1.  1.]
   weighted sum = xi*xib + xi/xib + xib/xi = C_UV = P_gg/(2 Nc)   (checked at xi=0.3, 0.65)

E. UV coefficient of the xi-integrated kernel, and how it splits over the blocks
   kappa=0.5 : r^2<G> / [(s.P)/(P^2 s^2)] = +39.613189   predicted 2 l_k - 11/6 = +39.613198
             l_k block +2.000000 (=2)   ln(B/A)+rational blocks -1.833335 (=-11/6)
   kappa=1.0 : r^2<G> / [(s.P)/(P^2 s^2)] = +39.613182   predicted 2 l_k - 11/6 = +39.613198
             l_k block +1.999999 (=2)   ln(B/A)+rational blocks -1.833334 (=-11/6)

F. the xi-integrals of C_UV  (mpmath, 30 digits, lambda=1e-9)  -- the paper's (C.10)
   int C_UV dxi              = 39.613198342559     2 l_k - 11/6          = 39.613198340559
   int C_UV ln(xi xib) dxi   = -429.02139290167    -l_k^2 + 67/18 - pi^2/3 = -429.02139286022
   int C_UV ln(1/xib) dxi    = 214.51069645084     l_k^2/2 + pi^2/6 - 67/36 = 214.51069643011
   consistency  I[ln(xi xib)] + 2 I[ln(1/xib)] = -7.6498848e-22

G. regularity of each J_i as C = B - A -> 0  (spurious 1/C, 1/C^2, 1/C^3 poles cancel)
   C/B= 1.0e-02 :  J1*B=0.16750503 (1/6)  J3*B=0.50335854 (1/2)  J5*B=0.50167505 (1/2)  J6*B=1.00503359 (1)
                 J2*B-l_k=-0.99498325 (-1)   J4*B-l_k=-0.80585952 (-1)
   C/B= 1.0e-04 :  J1*B=0.16672983 (1/6)  J3*B=0.50003333 (1/2)  J5*B=0.50001667 (1/2)  J6*B=1.00005000 (1)
                 J2*B-l_k=-0.99995000 (-1)   J4*B-l_k=-0.99807748 (-1)
   C/B= 1.0e-06 :  J1*B=65.98959961 (1/6)  J3*B=0.49993451 (1/2)  J5*B=0.50006599 (1/2)  J6*B=1.00000050 (1)
                 J2*B-l_k=-0.99999950 (-1)   J4*B-l_k=-0.99998078 (-1)
```

Code

`zint/src/xi_integration.py` — symbolic check that the six V_i with weights $(\xi\bar{\xi}, \xi/\bar{\xi}, \bar{\xi}, \bar{\xi}/\xi, \xi, 1)$ reproduce the contraction of (1.1) (true for $\kappa = \frac{1}{2}$, false for $\kappa = 1$). `zint/src/xi_masters.py` — the six J_i. `zint/src/xi_assemble.py` — $\mathcal{G}(r)$, the three blocks, the UV separation. `zint/src/run_xi.py` — every number above, written to `zint/RESULTS_xi.txt`.